

Query

Given vectors $\vec{a} = (1,2)$ and $\vec{b} = (-3,2)$, find the real number k such that $k\vec{a} + \vec{b}$ is in the same or opposite direction as $\vec{a} - 3\vec{b}$.

Reasoning Summary

I'll break down the steps clearly: first, I'll compute k as $a + b$. Then, I'll compute $a - 3b$. After that, I need to set the proportional relationship. Finally, I'll solve the resulting equations. I want to make sure each step is clear and makes sense. Once I have everything laid out, I can produce the final answers. That way, I can ensure I'm covering everything properly!

Final Answer

Let $\vec{a} = (1,2)$, $\vec{b} = (-3,2)$. Then $k\vec{a} + \vec{b} = (k - 3, 2k + 2)$, $\vec{a} - 3\vec{b} = (10, -4)$

For these vectors to be parallel, there exists $t \in \mathbb{R}$ such that $(k - 3, 2k + 2) = t(10, -4)$.

Hence $\frac{k-3}{10} = \frac{2k+2}{-4}$. Solving, $-4(k - 3) = 10(2k + 2) \Rightarrow -4k + 12 = 20k + 20 \Rightarrow -24k = 8 \Rightarrow k = -\frac{1}{3}$.

Thus, $k = -\frac{1}{3}$.

Synthesized Reasoning Trace

Okay, let's see. I need to find the real number k such that when I take k times vector a plus vector b , the result is either in the same direction or exactly opposite to vector a minus 3 times vector b . Hmm, directions here matter. So, vectors being in the same or opposite direction means they're scalar multiples of each other, right? So, maybe I can set up an equation where one vector is a scalar multiple of the other. First, let me write down the vectors. Vector a is $\{1, 2\}$, and vector b is $\{-3, 2\}$. The problem says $k*a + b$ should be parallel (same or opposite direction) to $a - 3*b$. For two vectors to be parallel, their components must be proportional. That is, there exists some scalar λ such that $k*a + b = \lambda*(a - 3b)$. So, maybe I can set up equations based on that proportionality.

[... reasoning steps omitted for space ...]

Alternatively, perhaps the vectors could be in the same direction, which would mean λ positive. But in this case, since we found a solution with λ negative, that's valid. So unless there's another constraint, this should be the answer.

Wait, let me check if there are other possibilities. Suppose the vectors are scalar multiples, but maybe with different scalars. Wait, no. Because if they are in the same direction, then λ is positive, and opposite direction would be λ negative. So, the only solutions are when the vectors are scalar multiples, regardless of the sign. So, even if it's negative, it's still a valid scalar multiple. So, our previous solution is correct.

Therefore, the value of k is $-1/3$.